

Shengqi Cui

Great Neck, NY • sc2932@cornell.edu • (516) 232-7862

EDUCATION

Cornell University, College of Engineering

B.S. in Biomedical Engineering (Biomedical Imaging and Instrumentation Concentration)

GPA: 3.89

Ithaca, NY

May 2027 (expected)

Cornell University, College of Engineering

Masters of Engineering in Electrical & Computer Engineering

Ithaca, NY

December 2027 (expected)

RELEVANT COURSEWORK

BME 3030: **Biomedical Circuit, signals and systems** | ECE 3150: **Microelectronics** | BME 3310: **Medical Imaging** | ECE 4530:

Analog Integrated Circuits design | ECE 4330: **Microwave circuit design**

PROJECTS

Lumikey | Founder

Ithaca, NY

July 2025 – Present

- Designed and built a modular LED piano guidance product with custom 4-layer PCB, ESP32-S3, IR detection, and wireless multi-module communication via ESP-NOW.
- Working with Cornell eLab accelerator towards commercialization.

Fandia | Electrical R&D

Ithaca, NY

February 2026 – Present

- Designed the PCB and prototyped the second generation of a modular benchtop bioreactor. Received interest from 6 research labs.
- Designing an electrowetting-on-dielectric liquid handling system for automating downstream analysis processes.

Cornell Biomedical Device | Project manager/New member education Chair

Ithaca, NY

September 2025 – Present

- Leading a team of six members on building a wearable biomechanical monitoring device to detect abnormal joint hyperextension, heart rate, blood flow in patients with EDS (Ehlers-Danlos syndrome) and provide a holistic data collection system for patients with EDS.
- Developed and led a new member education program incorporating breadboarding, PCB design, CAD software and 3D printing and a final project.

RESEARCH EXPERIENCE

Undergraduate Research Assistant

Ithaca, NY

Adie Lab, Cornell University

Jan 2026 – Present

- Achieved an order of magnitude speed up using GPU-accelerated CUDA pipeline for resolution enhanced OCT image processing.
- Preparing presentation in SPIE Photonics West.

Undergraduate Research Assistant

Ithaca, NY

Lee Lab, Cornell University

December 2023 – May 2024

- Investigated the mechanisms of Triple Negative Breast Tumor Cell (MDA-MB-231) on the morphology and junction of engineered lymphatic vessels using 3-D tumor-on-a-chip
- Trained on cell culturing, cell passaging, drainage assay, confocal imaging, fabrication of vessel organ-on-chips

LEADERSHIP AND EXTRACURRICULARS

Teaching Assistant

Ithaca, NY

Cornell University

August 2025 – Present

- TA for BME 2110 (Biomolecular Thermodynamics), BME 3030 (Biomedical Circuit, signals and systems), and BME 3020 (Molecular Principles of BME): lead weekly discussion and exam review sessions, hold office hours, administer oral and written quizzes, and assist with grading and lab activities for undergraduate and graduate students

Real A Capella

Ithaca, NY

Baritone Singer, Music Director

August 2023 – Present

- Practiced and performed Chinese Pop songs in collaboration with other vocal parts
- Arranged pieces for acapella and directs 2-hour weekly rehearsals and musical performance

SKILLS

Technical skills: C/C++ • Python • Fusion 360 • Altium • Keysight ADS • Cadence Virtuoso

Interests: Snowboarding • Piano • Singing • Saxophone • Fishing • Cooking

Language: English (Native) • Mandarin (Native) • French (intermediate)